

Introduction to High-Performance and Parallel...  
Computing

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Today's goals



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Module 1

High-Performance Computing (HPC) for Non-Computer Scientists

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Course Overview and Syllabus

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Course Updates and Accessibility Support

Reading • 1 min

Course Overview

Video • 2 min

✓

Earn Academic Credit for your Work!

Reading • 10 min

✓

Course Support

Reading • 10 min

Course Syllabus

Reading • 10 min

Introduce Yourself

Discussion Prompt • 10 min

Technical Help and Discussions

Technical Help and Discussions

Discussion Prompt • 10 min

JupyterLab and Submitting Assignments

Tour of JupyterLab

Video • 4 min

Submitting Assignments

Video • 6 min

Practice Assignment Submission

Practice Assignment Submission

Programming Assignment • 30 min

Introduction to Linux

Linux - Part 1

Video • 5 min

Linux - Part 2

Video • 3 min

Accessing Remote Systems

Accessing Remote Systems

Video • 6 min

Filesystems

Filesystems

Video • 4 min

Bash Scripting

Bash Scripting, Part 1

Video • 7 min

Bash Scripting - Part 2

Video • 5 min

Module Assignments

Week 1 Quiz

Graded Assignment • 30 min

Bash Scripting

Programming Assignment • 2h

Module 2

Nuts and Bolts of HPC

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Module 3

Basic Parallelism

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Module 4

Evaluating Parallel Program Performance

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# Course Syllabus

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## Instructors

Shelley Knuth  
Thomas Hauser

## Course Description

This course introduces the fundamentals of high-performance and parallel computing. It is targeted toward scientists, engineers, scholars, or anyone seeking to develop the software skills necessary for work in parallel software environments. These skills include big-data analysis, machine learning, parallel programming, and optimization. We will cover the basics of Linux environments and bash scripting all the way to high throughput computing and parallelizing code.

## Course Objectives

After completing this course, you will be familiar with:

- The components of a high-performance distributed computing system
- Types of parallel programming models and the situations in which they might be used
- High-throughput computing
- Shared memory parallelism
- Distributed memory parallelism
- Navigating a typical Linux-based HPC environment
- Assessing and analyzing application scalability including weak and strong scaling
- Quantifying the processing, data, and cost requirements for a computational project or workflow

## Prerequisites

General experience using a computer is required. Basic knowledge of Linux or how to navigate a filesystem is helpful but not required. A high level of understanding of the role of supercomputers or large-scale computing is also helpful but not required. Finally, familiarity with either Fortran 90, C++, or Python will be helpful in completing the programming assignments.

## Grade Requirements

| Assignment                                                  | Percentage of Grade |
|-------------------------------------------------------------|---------------------|
| Quiz: Week 1 Quiz                                           | 5%                  |
| Programming Assignment: Practice Assignment Submission      | 1%                  |
| Programming Assignment: Bash Scripting                      | 15%                 |
| Quiz: Week 2 Quiz                                           | 5%                  |
| Programming Assignment: Submitting Jobs with Slurm - Part 1 | 7.5%                |
| Programming Assignment: Submitting Jobs with Slurm - Part 2 | 7.5%                |
| Quiz: Week 3 Quiz                                           | 5%                  |
| Programming Assignment: Simple Application Timing           | 19%                 |
| Quiz: Week 4 Quiz                                           | 5%                  |
| Programming Assignment: Strong Scaling Study                | 15%                 |
| Programming Assignment: Weak Scaling Study                  | 15%                 |

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